

Map Descriptive Analysis

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```
import geopandas as gpd
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import os

out_dir = "visualizations/descriptive_maps"
os.makedirs(out_dir, exist_ok=True)

from CESTools.utils import clean_data
from CESTools.utils import qmap

shp_path = "data/calenviroscreen40_shp/CES4 Final Shapefile.shp"
shape = gpd.read_file(shp_path)

shape = clean_data(shape, keep_geom=True)
shape = shape.to_crs(epsg=3857)

shape.head()
```

	CIscoreP	OzoneP	PM2_5_P	DieselPM_P	TrafficP	Tox_Rel_P	PesticideP	Hispanic	White	AfricanAm	NativeAm	OtherMult	AAPI	Geometry
0	69.162585	273.402234	483.700065	208.0672	68.369776	828.92368	990.267912	0.090909	0.480909	0.090909	0.090909	0.090909	0.090909	POLYGON ((- 13406867.885 4155533.339, - 13404832...
1	70.637902	273.402234	483.700065	208.0672	68.369776	828.92368	990.267912	0.090909	0.480909	0.090909	0.090909	0.090909	0.090909	POLYGON ((- 13406867.885 4155533.339, - 13406867...
2	61.069086	273.402234	483.700065	208.0672	68.369776	828.92368	990.267912	0.090909	0.480909	0.090909	0.090909	0.090909	0.090909	POLYGON ((- 13404824.281 4156507.571, - 13404832...
3	5.988106	632.395071	869.062659	07.050606	8.355011	782.96879	989.171253	6690	0	0	0	0	0	POLYGON ((- 13403841.327 4147463.306, - 13403732...
4	23.123163	692.733071	324.202559	01.070809	2.302002	385.88878	698.472372	3680	0	0	0	0	0	POLYGON ((- 13406428.948 4147489.901, - 13404941...

5 rows × 31 columns

```
burden = "CIscoreP"
```

```
exposure_vars = ["OzoneP", "PM2_5_P", "DieselPM_P", "TrafficP", "Tox_Rel_P", "PesticideP"]
shape["ExposureIdx"] = shape[exposure_vars].mean(axis=1, skipna=True)
```

```
race_cols = ['Hispanic', 'White', 'AfricanAm', 'NativeAm', 'OtherMult', 'AAPI']
race_bins = [5, 10, 20, 40, 60, 80, 100]
```

```
for race in race_cols:
    print(f" - - - {race} - - -")
```

```
# 1: Map depicting race share (for each race)
qmap(
```

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        shape,
        race,
        title=f"{race} share (fixed bins)",
        cmap="plasma",
        bins=race_bins,
        save=f"visualizations/descriptive_maps/race_{race}_fixedbins.png"
    )

# 2) Bivariate map: race share (x) vs burden (y), 3 \times 3 quantiles

df = shape[[race, burden, "geometry"]].dropna().copy()

# quantile bins: 0 = low, 1 = mid, 2 = high
df["race_bin"] = pd.qcut(df[race], 3, labels=False, duplicates="drop")
df["burden_bin"] = pd.qcut(df[burden], 3, labels=False, duplicates="drop")

df["bi"] = df["race_bin"].astype(int) + 3 * df["burden_bin"].astype(int)

# rows = burden (low to high), cols = race (low to high)
colors = [
    "#e8e8e8", "#b5c0da", "#6c83b5", # low burden
    "#b8d6be", "#90b2b3", "#567994", # mid burden
    "#73ae80", "#5a9178", "#2a5a5b" # high burden
]
df["color"] = df["bi"].map(lambda i: colors[int(i)])

fig, ax = plt.subplots(1, 1, figsize=(10,10))
ax.set_axis_off()

# base map
shape.plot(ax=ax, color="lightgrey", edgecolor="white", linewidth=0.05)
df.plot(ax=ax, color=df["color"], edgecolor="white", linewidth=0.05)

ax.set_title(
    f"Bivariate: {race} share ( \rightarrow) vs {burden} (↑), 3 \times 3 quantiles",
    fontsize=14
)

# legend (3 \times 3 grid)
leg = fig.add_axes([0.67, 0.12, 0.22, 0.22])
leg.set_xticks([]); leg.set_yticks([]); leg.set_frame_on(True)

k = 0
for j in range(3): # burden: low -> high
    for i in range(3): # race: low -> high
        leg.add_patch(

```

```

        plt.Rectangle((i, j), 1, 1, facecolor=colors[k], edgecolor="none")
    )
    k += 1

leg.set_xlim(0, 3); leg.set_ylim(0, 3)
leg.set_xlabel(f"{race} \rightarrow", fontsize=9)
leg.set_ylabel(f"{burden} \rightarrow", fontsize=9)
plt.savefig(
    f"visualizations/descriptive_maps/bivariate_{race}_{burden}.png",
    dpi=300,
    bbox_inches="tight"
)
plt.show()

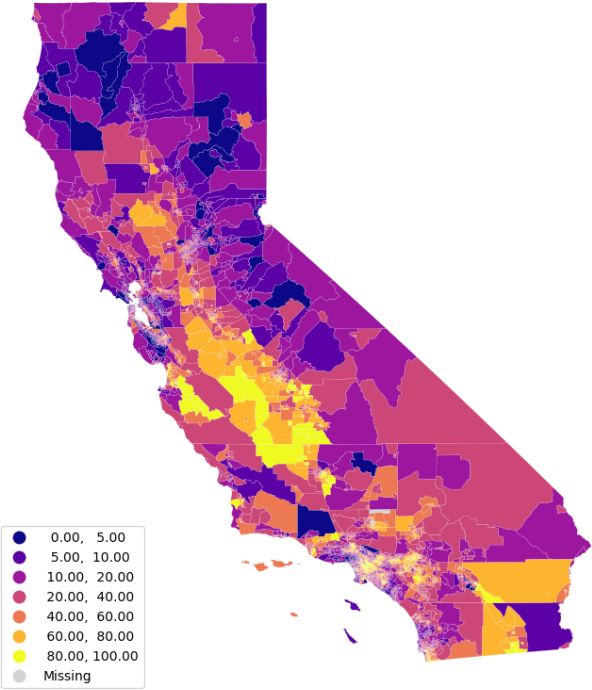
# 3) Overlap hotspots: top 10% burden AND top 20% race share
cut_hi = shape[race].quantile(0.80)
burden_hot = shape[burden].quantile(0.90)
overlap = (shape[burden] >= burden_hot) & (shape[race] >= cut_hi)

fig, ax = plt.subplots(1, 1, figsize=(10,10))
ax.set_axis_off()
shape.plot(ax=ax, color="lightgrey", edgecolor="white", linewidth=0.05)
shape.loc[overlap].plot(ax=ax, color="crimson", edgecolor="white", linewidth=0.05)
ax.set_title(f"Overlap hotspots: top 10% {burden} AND top 20% {race} share", fontsize=14)
plt.savefig(
    f"visualizations/descriptive_maps/overlap_hotspots_{race}_{burden}.png",
    dpi=300,
    bbox_inches="tight"
)
plt.show()

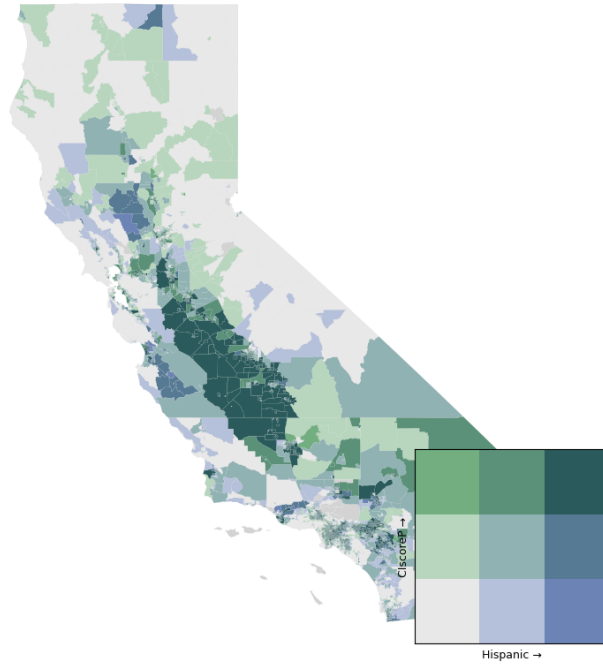
- - - Hispanic - - -

```

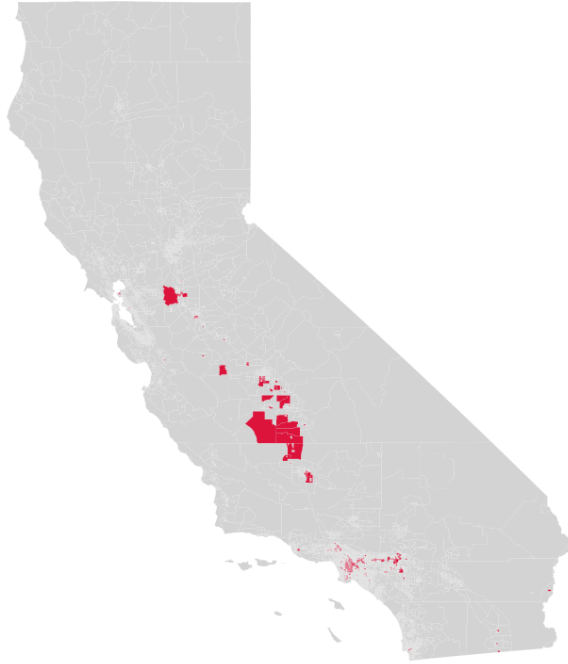
Hispanic share (fixed bins)



Bivariate: Hispanic share (→) vs ClscoreP (↑), 3×3 quantiles

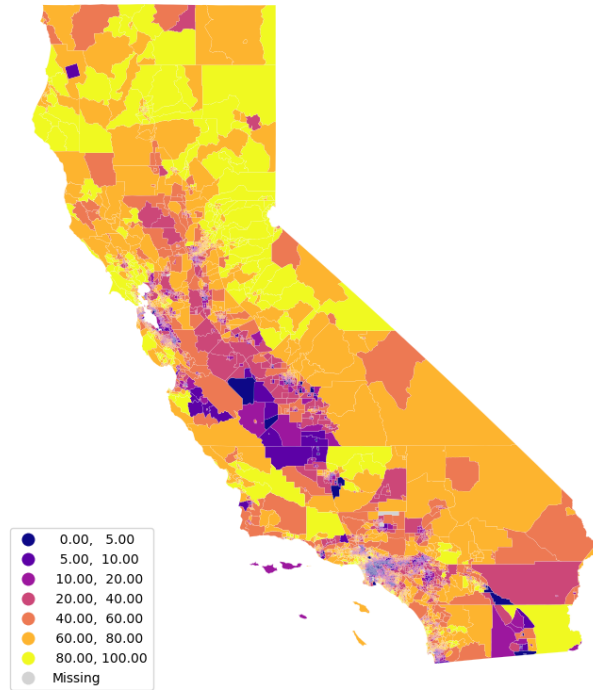


Overlap hotspots: top 10% CiscoreP AND top 20% Hispanic share

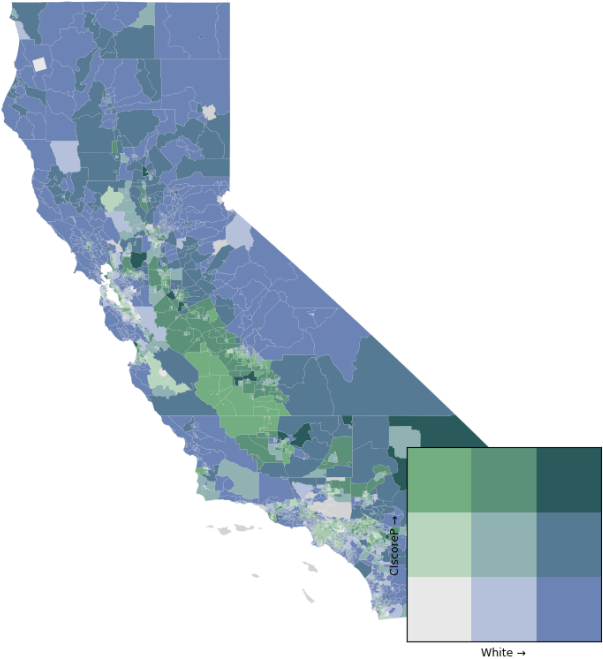


- - - White - - -

White share (fixed bins)



Bivariate: White share (→) vs CiscoreP (↑), 3×3 quantiles

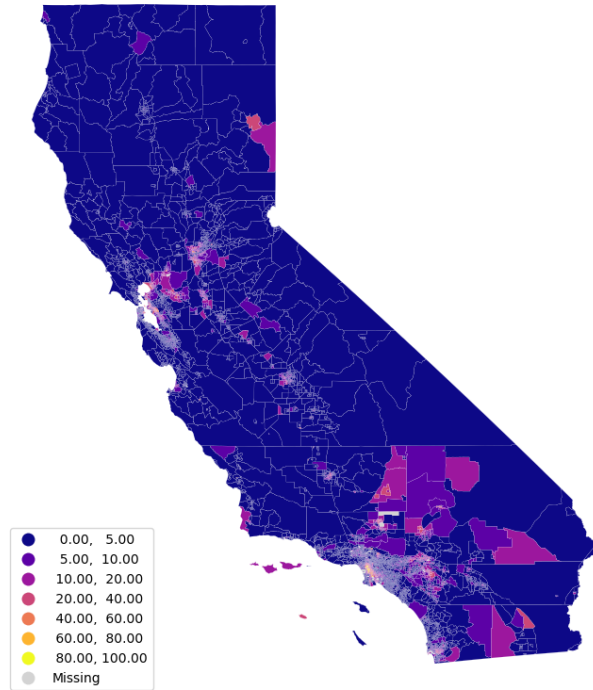


Overlap hotspots: top 10% CIscoreP AND top 20% White share

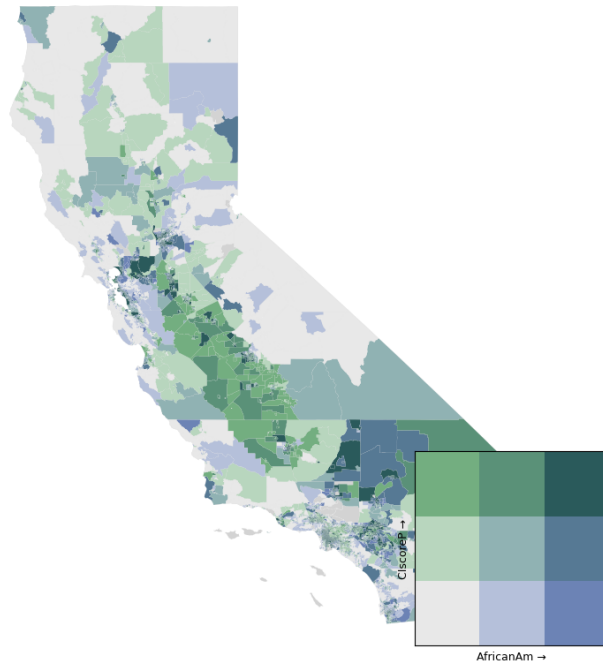


- - - AfricanAm - - -

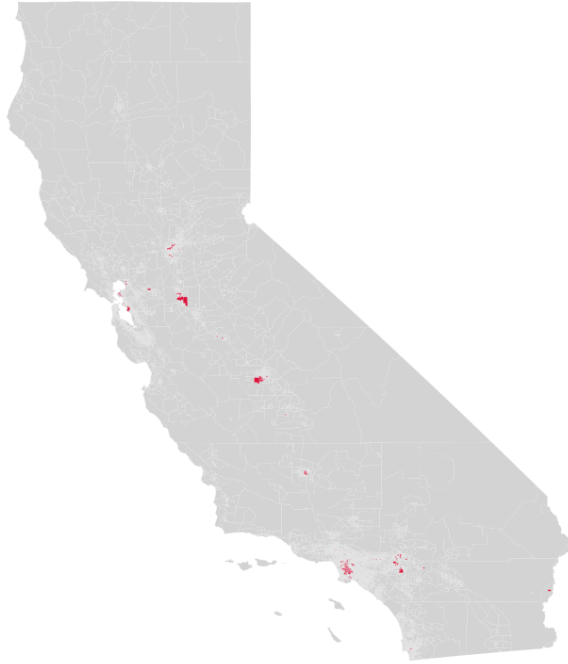
AfricanAm share (fixed bins)



Bivariate: AfricanAm share (→) vs CIscoreP (↑), 3×3 quantiles

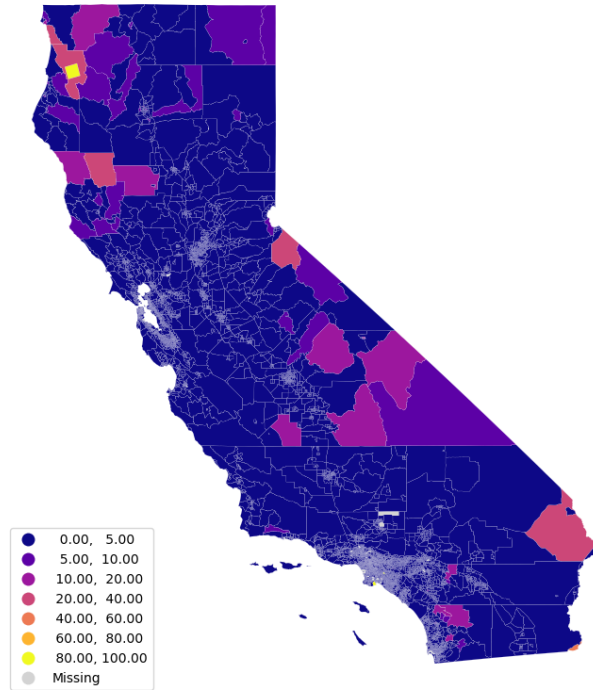


Overlap hotspots: top 10% CIscoreP AND top 20% AfricanAm share

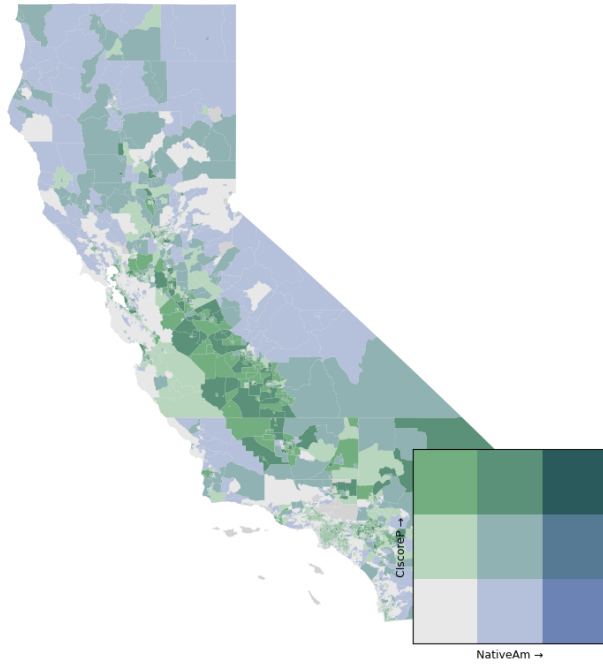


- - - NativeAm - - -

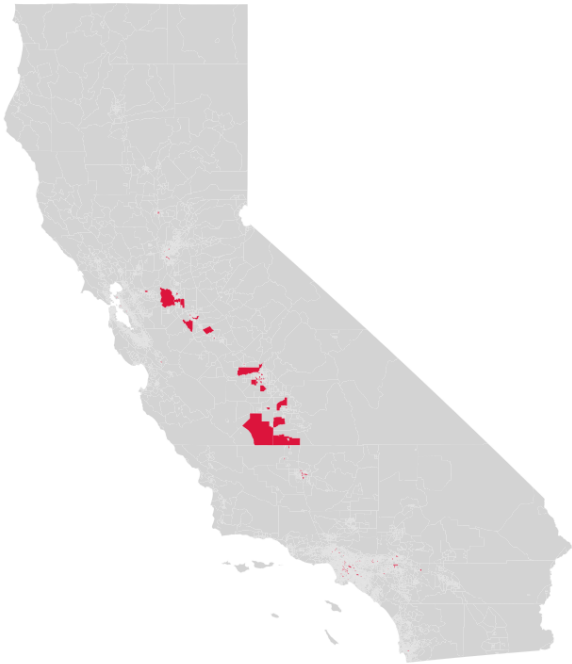
NativeAm share (fixed bins)



Bivariate: NativeAm share (→) vs ClscoreP (↑), 3×3 quantiles

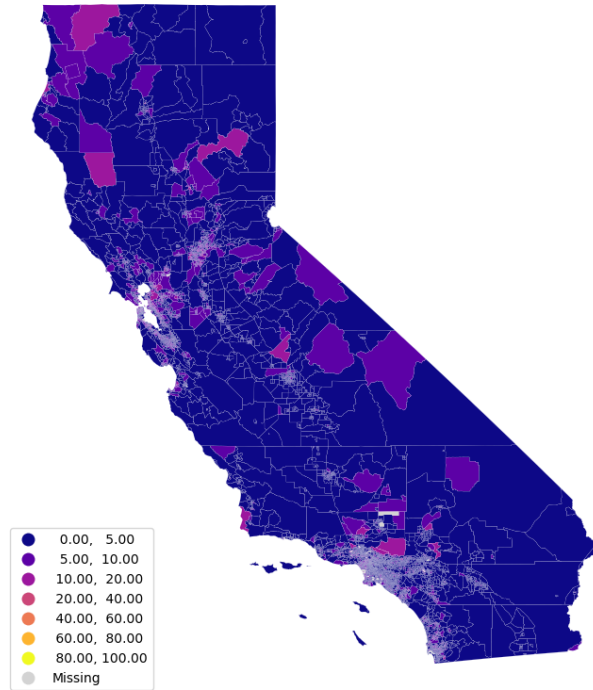


Overlap hotspots: top 10% CIscoreP AND top 20% NativeAm share

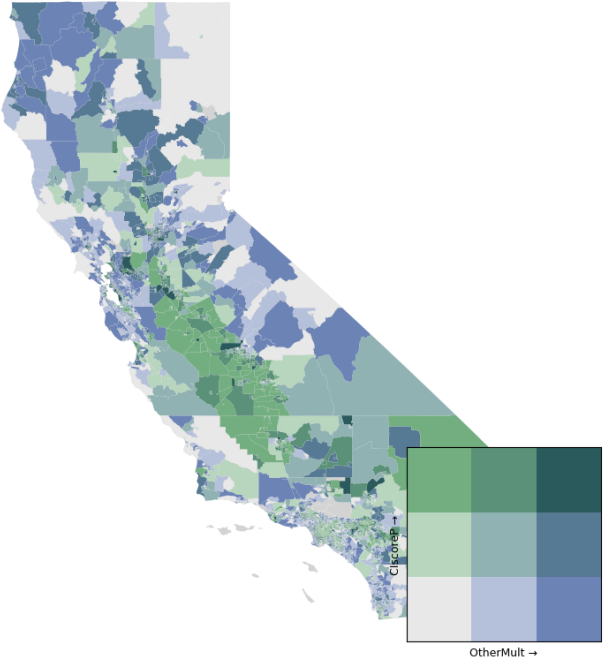


- - - OtherMult - - -

OtherMult share (fixed bins)



Bivariate: OtherMult share (→) vs CIscoreP (↑), 3×3 quantiles

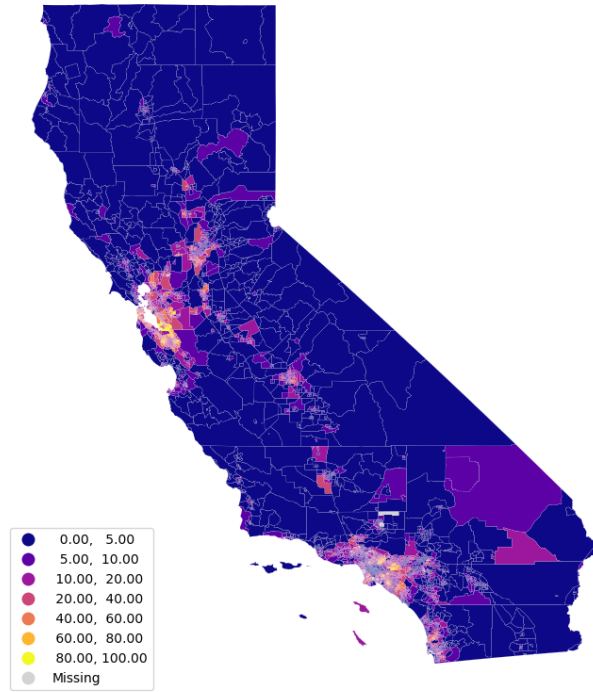


Overlap hotspots: top 10% CIscoreP AND top 20% OtherMult share

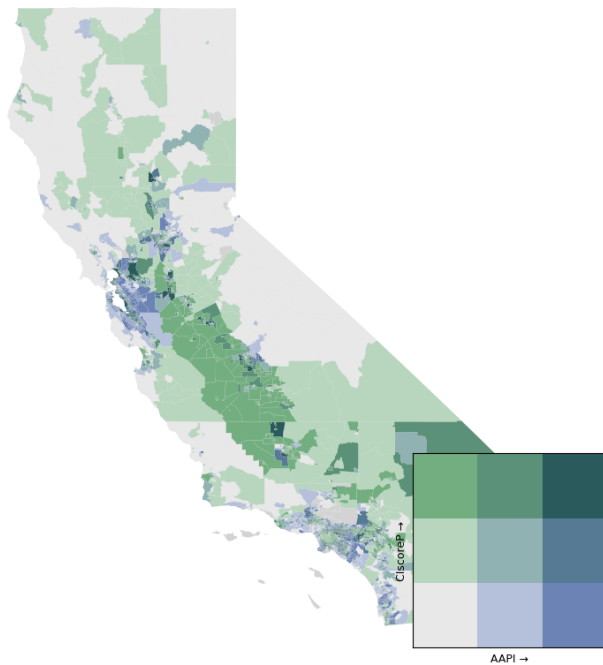


- - - AAPI - - -

AAPL share (fixed bins)



Bivariate: AAPI share (→) vs ClscoreP (↑), 3×3 quantiles



Overlap hotspots: top 10% CiscorP AND top 20% AAPI share

